

\* Results:  $S = \int_M L + \int_{\partial M} \ell$

• They don't consider null boundaries...

$$\delta L = E_a \delta \phi^a + d\Theta, \quad E_a = 0 \text{ (E.o.M)}$$

$\Theta$ :  $(d-1)$ -form, linear in  $\delta \phi^a$

$$(\Theta + \delta L)_\Gamma = dC$$

$\Gamma$ : Spatial boundary

$C$ :  $(d-2)$  form on  $\Gamma$ , linear in  $\phi$ .

• (Pre) symplectic form of this system is given by

$$\tilde{\Omega} = \int_\Sigma \delta(\Theta - dC)$$

•  $\xi^\mu$ : vect. field diffeo

$$H_\xi = \int_\Sigma \mathcal{H}_\xi + \int_{\partial \Sigma} (\xi \cdot \ell - X_\xi \cdot C)$$

Notation:  $(*w)_{\mu_1 \dots \mu_{d-p}} = \frac{1}{p!} \epsilon^{\nu_1 \dots \nu_p}_{\mu_1 \dots \mu_{d-p}} w_{\nu_1 \dots \nu_p}$

$$\mathcal{L}_X w = i_X(dw) + d(i_X w)$$

Here they use  $X \lrcorner \alpha =: i_X(\alpha)$

• For  $w = \frac{1}{p!} w_{\mu_1 \dots \mu_p} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}$

$$i_\xi w = \frac{1}{(p+1)!} \xi^\mu w_{\mu \mu_1 \dots \mu_p} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}$$

• Can write  $i_\xi$  as a Grassmann derivative...

$$i_\xi = \xi^\mu \frac{\partial}{\partial (dx^\mu)}$$

• If  $u \in \Omega^0(M)$  then from the above we simply have  $i_\xi(u) = 0$ .

• Given an orientation on  $M$ ,  $\exists$  a natural induced orientation of  $\partial M$  s.t.

$$\int_M dw = \int_{\partial M} w$$

• Regime  $E_{\text{on}}$  is related to the spacetime volume by

$$E = M \wedge E_{\text{on}} \quad \partial \Gamma$$

$n$ : outward pointing

$$n_\mu \equiv \frac{1}{\sqrt{|\partial_\alpha f \partial_\beta f g^{\alpha\beta}|}}$$

\* Orientation on  $\Sigma$  given by viewing it as the boundary of its part in  $M$

ex:  $M: \epsilon = dt \wedge dx \wedge dy \wedge dz \mid x \leq 0$

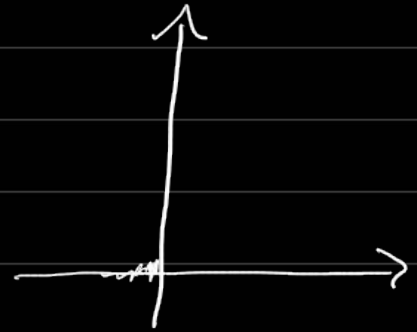
$\Sigma$ : surface  $t=0$ ,  $n_\mu = (0, 1, 0, 0)$

$n = dx$

$\epsilon_{\partial\mu} = -dt \wedge dy \wedge dz$

$\epsilon_\Sigma = dx \wedge dy \wedge dz$

$\epsilon_{\partial z} = dy \wedge dz$



Formalism:  $\dot{q}^\alpha = \frac{\partial H}{\partial p^\alpha}$ ,  $\dot{p}^\alpha = -\frac{\partial H}{\partial q^\alpha}$

$\Omega$ : Symplectic form on  $P \leftarrow$  phase space

$\Omega: TM \rightarrow T^*M$

$T^*M \ni \Omega(Y)[x] = \Omega(Y, x)$

$\Rightarrow \Omega^{-1}(\omega, \sigma) \equiv \omega(\Omega^{-1}(\sigma))$ ,  $\omega, \sigma \in T^*M$

Given  $H: P \rightarrow \mathbb{R}$ , we can define  $X_H$  on  $P$

$X_H(f) := \Omega^{-1}(\delta f, \delta H)$

$f: P \rightarrow \mathbb{R}$

$\{f, g\} \equiv \Omega^{-1}(\delta f, \delta g) = \Omega(X_g, X_f)$

(Poisson Bracket)

$X_H(f) \equiv \Omega(X_H, X_f) = \frac{\partial f^\mu}{\partial x^\mu} \frac{\partial H^\nu}{\partial \sigma} \Omega(\partial_\nu, \partial_\mu)$

$\dot{f} = \frac{df}{dt} = \frac{\partial f}{\partial x^\mu} \frac{\partial x^\mu}{\partial t}$

$\dot{f} \equiv X_H(f) = \{f, H\}$

$i_{X_H} \Omega = -\delta H$

$= (X_H)^\mu \Omega_{\mu\nu} dx^\nu$

$(\delta H)_\nu = (X_H)^\mu \Omega_{\mu\nu}$

\* For  $\Omega = \sum_a \delta p^a \wedge \delta q^a$

\* View set of  $sol^n$  of E.C.M. as pre-phase space

$\tilde{P}$  and  $P$  is obtained by an appropriate quotient

$\tilde{\Omega}^{(2)} \in \mathcal{P} \subset \mathcal{P}$   $\tilde{\mathcal{P}}/\sim \sim \mathcal{P}$   
 $\tilde{\Omega}$ : Pre-symplectic form on  $\tilde{\mathcal{P}}$   
 $d\tilde{\Omega}^{(2)}$

•  $\tilde{X}, \tilde{Y} \in T\tilde{\mathcal{P}}$ , then s.t.  $0 = \tilde{\Omega}_{XX} = \tilde{\Omega}_{XX} = \tilde{\Omega}_{YY}$

$$[\tilde{X}, \tilde{Y}] = \mathcal{L}_{\tilde{X}} \tilde{Y}$$

will also be annihilated by  $\tilde{\Omega}$ :

proof: If  $\tilde{\Omega}(\tilde{X}, \tilde{X}) = 0 = \tilde{\Omega}(\tilde{X}, \tilde{Y})$

$$\begin{aligned} \tilde{\Omega}(\mathcal{L}_{\tilde{X}} \tilde{Y}, \mathcal{L}_{\tilde{X}} \tilde{Y}) &= \\ (\tilde{X}^\mu \partial_\mu \tilde{Y}^\nu) \partial_\nu &- (\tilde{Y}^\mu \partial_\mu \tilde{X}^\nu) \partial_\nu \\ &= (\tilde{X}^\mu \partial_\mu \tilde{Y}^\nu - \tilde{Y}^\mu \partial_\mu \tilde{X}^\nu) \partial_\nu \end{aligned}$$

$$\text{But } \tilde{X}^\mu \tilde{\Omega}_{\mu\nu} = \tilde{Y}^\mu \tilde{\Omega}_{\mu\nu} = 0$$

$$\begin{aligned} &(\tilde{X}^\mu \partial_\mu \tilde{Y}^\nu - \tilde{Y}^\mu \partial_\mu \tilde{X}^\nu) \tilde{\Omega}_{\nu\sigma} \\ &= \partial_\mu (\tilde{X}^\mu \tilde{Y}^\nu \tilde{\Omega}_{\nu\sigma}) - \partial_\mu (\tilde{Y}^\mu \tilde{X}^\nu \tilde{\Omega}_{\nu\sigma}) - \tilde{X}^\mu \tilde{X}^\nu \partial_\mu \tilde{\Omega}_{\nu\sigma} \\ &\quad - \partial_\mu (\tilde{X}^\nu \tilde{Y}^\mu \tilde{\Omega}_{\nu\sigma}) + \partial_\mu \tilde{Y}^\mu \tilde{X}^\nu \tilde{\Omega}_{\nu\sigma} + \tilde{X}^\mu \tilde{Y}^\mu \partial_\mu \tilde{\Omega}_{\nu\sigma} \\ &= 0 \quad \checkmark \end{aligned}$$

So  $[\tilde{X}, \tilde{Y}]$  is also eliminated by  $\tilde{\Omega}$

• Note:  $\tilde{X}^\mu \tilde{\Omega}_{\mu\nu} = \tilde{Y}^\mu \tilde{\Omega}_{\mu\nu} = 0$  is simply

$$\begin{aligned} i_X \tilde{\Omega} &= i_Y \tilde{\Omega} = 0 \\ \mathcal{L}_X \tilde{\Omega} &= 0 \end{aligned}$$

$$\text{Claim: } \mathcal{L}_{\tilde{X}}(i_{\tilde{Y}} \tilde{\Omega}) = i_{\mathcal{L}_{\tilde{X}} \tilde{Y}} \tilde{\Omega} + i_{\tilde{Y}}(\mathcal{L}_{\tilde{X}} \tilde{\Omega})$$

$$\begin{aligned} \hookrightarrow \mathcal{L}_{\tilde{X}}(\tilde{Y}^\alpha \tilde{\Omega}_{\alpha\beta} dx^\beta) &= \mathcal{L}_{\tilde{X}}(\tilde{Y}^\alpha \tilde{\Omega}_{\alpha\beta}) dx^\beta + \tilde{Y}^\alpha \tilde{\Omega}_{\alpha\beta} \mathcal{L}_{\tilde{X}}(dx^\beta) \\ &= \tilde{X}^\mu \partial_\mu (\tilde{Y}^\alpha \tilde{\Omega}_{\alpha\beta}) dx^\beta + \tilde{Y}^\alpha \tilde{\Omega}_{\alpha\gamma} \partial_\beta \tilde{X}^\gamma dx^\beta \\ &= (\tilde{X}^\mu \partial_\mu (\tilde{Y}^\alpha \tilde{\Omega}_{\alpha\beta}) + \tilde{Y}^\alpha \tilde{\Omega}_{\alpha\gamma} \partial_\beta \tilde{X}^\gamma) dx^\beta \\ &\quad \tilde{X}^\mu \partial_\mu \tilde{Y}^\alpha \tilde{\Omega}_{\alpha\beta} + \tilde{X}^\mu \tilde{Y}^\alpha \partial_\mu \tilde{\Omega}_{\alpha\beta} + \tilde{Y}^\alpha \tilde{\Omega}_{\alpha\gamma} \partial_\beta \tilde{X}^\gamma \end{aligned}$$

$$\begin{aligned} \bullet \mathcal{L}_{[X, Y]} \tilde{\Omega} &= (\tilde{X}^\mu \partial_\mu \tilde{Y}^\nu - \tilde{Y}^\mu \partial_\mu \tilde{X}^\nu) \partial_\nu \tilde{\Omega}_{\alpha\beta} \\ &\quad + \partial_\alpha [\ ]^\sigma \tilde{\Omega}_{\sigma\beta} + \partial_\beta [\ ]^\sigma \tilde{\Omega}_{\alpha\sigma} \end{aligned}$$

$$\bullet i_{\tilde{Y}}(\mathcal{L}_{\tilde{X}} \tilde{\Omega}) = i_{\tilde{Y}}(\tilde{X}^\mu \partial_\mu + \dots)$$

• In their notation:

$$\begin{aligned} \mathcal{L}_{\tilde{X}} \tilde{Y} \cdot \tilde{\Omega} &= \mathcal{L}_{\tilde{X}}(\tilde{Y} \cdot \tilde{\Omega}) - \tilde{Y} \cdot \mathcal{L}_{\tilde{X}} \tilde{\Omega} \\ &= \tilde{Y} \cdot \mathcal{L}_{\tilde{X}} \tilde{\Omega} + \mathcal{L}_{\tilde{X}}(\tilde{Y} \cdot \tilde{\Omega}) - \tilde{Y} \cdot \mathcal{L}_{\tilde{X}} \tilde{\Omega} \end{aligned}$$

$$\begin{aligned}
 &= -\tilde{Y} \cdot \delta(\tilde{X} \cdot \tilde{\Omega}) + \delta(\tilde{X} \cdot \tilde{\Omega}) \cdot \tilde{Y} \\
 &= -\tilde{Y} \cdot [\tilde{X} \cdot \delta\tilde{\Omega} + \delta(\tilde{X} \cdot \tilde{\Omega})] \\
 &= 0 \quad (\text{trivially}).
 \end{aligned}$$

$$\tilde{G} \equiv \{ \tilde{X}, \tilde{Y} \in T\tilde{P} \mid \tilde{X} \cdot \Omega = \tilde{Y} \cdot \Omega = (d_x Y) \cdot \Omega = 0 \}$$

$\tilde{G}$ : set of diffeomorphisms for which  $\frac{d}{dt}$  generate  $\tilde{X}, \tilde{Y}$

$$P \equiv \tilde{P}/\tilde{G}$$

$$\pi: \tilde{P} \rightarrow P$$

$$p \rightarrow [p] \leftarrow G\text{-orbit in } P$$

$$\text{let } p \in P, X, Y \in T_p P, \exists q \in \tilde{P}, \tilde{X}, \tilde{Y} \in T_q \tilde{P}$$

$$\left\{ \begin{array}{l} X = \pi_* \tilde{X} \\ Y = \pi_* \tilde{Y} \end{array} \right\} \quad \Omega(X, Y) \equiv \pi^* \Omega(\tilde{X}, \tilde{Y})$$

$$\text{Maxwell theory: } L = -\frac{1}{2} F \wedge * F \quad | \quad d \wedge * B = B \wedge * d$$

$$\delta L = -\frac{1}{2} \delta(dA \wedge * dA)$$

$$= -\frac{1}{2} \delta(dA) \wedge * dA + (-\frac{1}{2}) d\delta A \wedge * dA$$

$$= -d(\delta A) \wedge * dA$$

$$= -d(\delta A \wedge * F) - \delta A \wedge d * dA$$

$$\delta L = -\delta A \wedge d * F - d(\delta A \wedge * F)$$

\* In terms of Pre symplectic Potential

$$\delta L = E(A) \wedge \delta A + d\Theta(A, \delta A)$$

$$E(A) = d * F = 0$$

$$\Rightarrow \Theta(A, \delta A) = -\delta A \wedge * F$$

$$\text{Symplectic Potential: } \omega \equiv \delta \Theta = \delta A \wedge * \delta F$$

• pre symplectic form

$$\tilde{\Omega} = \int_{\Sigma} \delta A \wedge * \delta F$$

$A$  and  $*F$  are canonical conjugates

• Zero modes:

$$X_{\lambda} \equiv \int d^4x \partial_{\mu} \lambda \frac{\delta}{\delta A_{\mu}}$$

$$X_{\lambda} \cdot \tilde{\Omega} = \int d^4x \partial_{\mu} \lambda \left( \frac{\delta}{\delta A_{\mu}(x)} \right) \int_{\Sigma} \delta A \wedge * \delta F$$

$$= \int_{\Sigma} d\lambda \wedge * dF = \int_{\Sigma} d(\lambda * \delta F)$$

$$= \int_{\partial \Sigma} \lambda * \delta F$$

$$\hookrightarrow \boxed{Q_{\lambda} \equiv \lambda \int * F}$$

